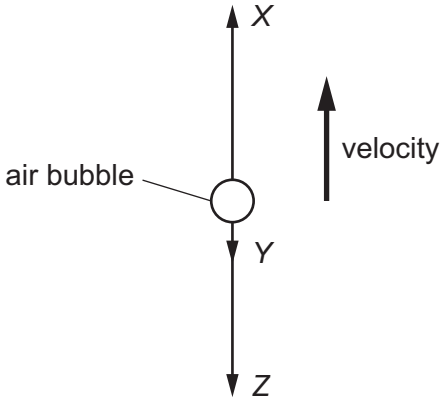
9 An air bubble in a tank of water is rising with terminal (constant) velocity. The forces acting on the bubble are *X*, *Y* and *Z*, as shown.



The upthrust on the bubble is *X*.

Which statement about the forces is correct?

A *Z* is the viscous force on the bubble, *Y* is the weight of the bubble and $X = Y + Z$.

B *Z* is the viscous force on the bubble, *Y* is the weight of the bubble and $X > Y + Z$.

C *Z* is the weight of the bubble, *Y* is the viscous force on the bubble and $X = Y + Z$.

D *Z* is the weight of the bubble, *Y* is the viscous force on the bubble and $X > Y + Z$.